

BE/APh161: Physical Biology of the Cell

Homework 4

Due Date: Wednesday, February 11, 2026

“We must travel in the direction of our fear.” - John Berryman

1. Estimating the diffusion constant.

In this problem, we are going to use the observed trajectories of diffusing GFP molecules to estimate the diffusion coefficient. You already did much of this in class.

(a) Conventional microscopy to observe individual fluorescent proteins moving freely in cytoplasm won't work. In this part of the problem, we are going to work out why. During a traditional experiment, the microscope shutter is open during some time interval of order 10s to 100s of milliseconds. By assuming a diffusion constant of $10 \mu\text{m}^2/\text{s}$, work out how far the fluorescent protein will move during the time that the shutter is open and compare that distance to the size of the cell itself and comment on how this limits our ability to measure the diffusion constant. Perform the estimate a second time, this time using the 0.3 ms exposure time shown in Figure 1(A).

(b) Using the trajectories shown in Figure 2 and our simple rule of thumb that $t_{\text{diffusion}} = L^2/D$ to estimate the diffusion constant for GFP. Explain your reasoning carefully.

2. Solving the Diffusion Equation

We have been interested in making estimates of the time scale associated with a number of biological processes. One of the most ubiquitous processes is diffusion, which is our “go to” null hypothesis for how molecules get around in cells.

In class I noted that the time scale for diffusing a distance L is given by $t = L^2/D$, where D is the diffusion constant. In this problem, we will formally derive this result. Note that parts (a) and (b) are effectively problem 13.2 of PBoC. Also, reading much of chap. 13 of PBoC will be very helpful

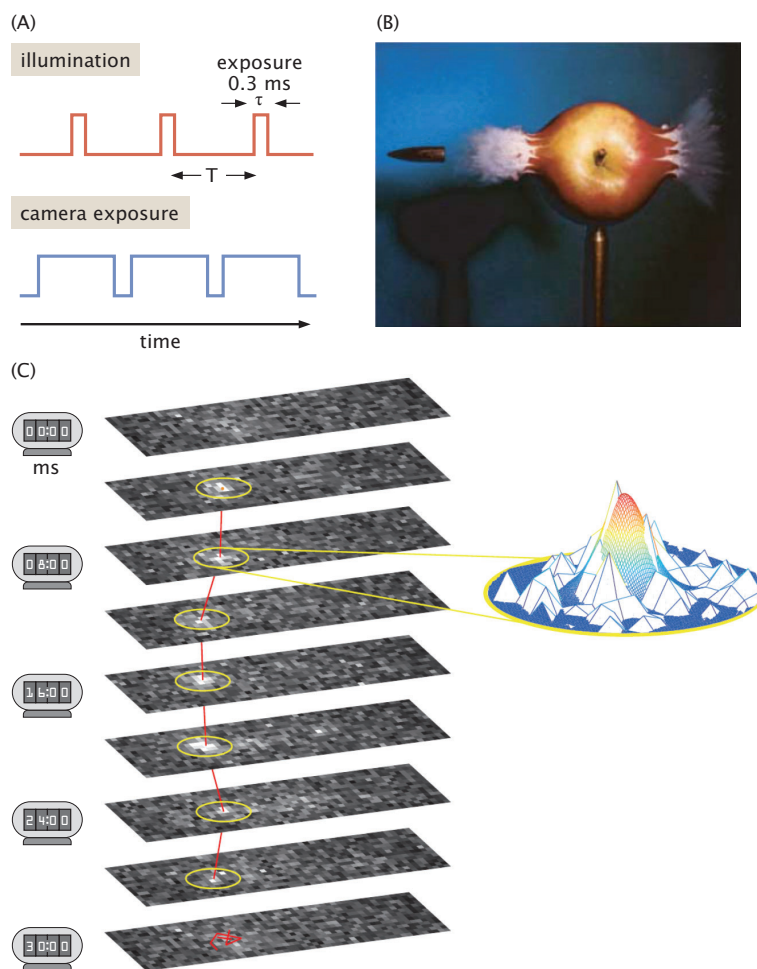


Figure 1: Stroboscopic illumination to capture fast protein dynamics. (A) By only illuminating a sample for a small fraction of the exposure time of a camera, it is possible to capture phenomena that would otherwise be blurred out. (B) A classic photo from MIT legend Harold Edgerton who pioneered stroboscopic photography for science and fun. Capturing the piercing of a bullet through an apple using stroboscopic illumination. (C) Measuring the position of an individual GFP molecule inside *E. coli*. (A, adapted from Harold and Esther Edgerton Foundation, 2006, courtesy of Palm Press, Inc.)

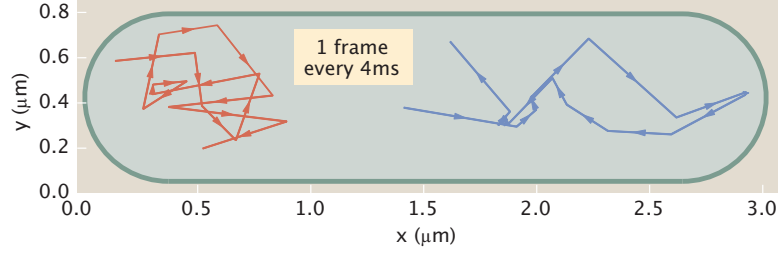


Figure 2: Time series showing positions of diffusing GFP molecules at different times. The red and blue traces correspond to different molecules. The lines are a guide to the eye. Adapted from BP English *et al.* Proc. Nat. Acad. Sci., 108:E365-E373, 2011.

for doing this problem.

(a) Our goal is to find the diffusive profile for some molecular species as a function of time. If we are given an initial concentration, we can use the diffusion equation to determine the concentration distribution at a later time. To that end, consider the one-dimensional diffusion equation in free space given by

$$\frac{\partial c(x, t)}{\partial t} = D \frac{\partial^2 c(x, t)}{\partial x^2}. \quad (1)$$

In particular, consider that the initial concentration distribution is given by $c(x, 0) = \delta(x)$, where $\delta(x)$ is the Dirac delta function and basically means that there is a spike at the origin. In particular, you will show that

$$G(x, t) = \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}, \quad (2)$$

where we introduce the Green function $G(x, t)$ to signify that this is the concentration profile for the special case in which the initial concentration is the spike at the origin as represented by the delta function.

To obtain the solution, we will Fourier transform the diffusion equation in the spatial variable x according to the Fourier transform convention

$$\tilde{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx, \quad (3)$$

and

$$f(x) = \int_{-\infty}^{\infty} \tilde{f}(k) e^{ikx} dk. \quad (4)$$

Using these definitions, Fourier transform both sides of the diffusion equation to arrive at the ordinary differential equation

$$\frac{d\tilde{c}(k, t)}{dt} = -Dk^2\tilde{c}(k, t). \quad (5)$$

Note that to do this Fourier transform of the right side, you will need to use integration by parts twice. Solve the differential equation that emerges from your Fourier transform to obtain $\tilde{c}(k, t)$ and make sure to use the initial condition $c(x, 0) = \delta(x)$ to find $\tilde{c}(k, 0)$. Then invert the Fourier transform on $\tilde{c}(k, t)$ to find $c(x, t)$. NOTE: You will need to use completion of the square to carry out the inversion. Make sure you explain all of your steps. We are big on having you not only do the analysis correctly, but also to explain what you are doing and why you are doing it. Also, explain why I said this is the solution for “free space”. Why would this solution fail to describe diffusion in a finite box?

(b) Using the solution we obtained above, find $\langle x \rangle$ and $\langle x^2 \rangle$. In general, we have that

$$\langle x^n \rangle = \frac{\int_{-\infty}^{\infty} x^n c(x, t) dx}{\int_{-\infty}^{\infty} c(x, t) dx}. \quad (6)$$

Explain what you find for both the first and second moments of the distribution as a function of time and explain how it relates to the estimated diffusion time $t = L^2/D$ which we use to find the time scale for diffusion over a length L . Using the Einstein-Stokes relation given by

$$D = \frac{k_B T}{6\pi\eta a}, \quad (7)$$

where η is the viscosity which for water is $\eta_{water} = 10^{-3} \text{ Pa s}$ and a is the radius of the diffusing particle, estimate the diffusion constant for a protein in water and make a log-log plot of diffusion time vs distance (with distances ranging from 1 nm to 100 mm) and comment on its biological significance. Also, make a plot of the solution for the point source as a function of time by showing $c(x, t)$ at various times t using the same diffusion constant.

(c) In their book “Cell Signaling”, Lim, Mayer and Pawson give the classic story about diffusion in neurons and how diffusion will take prohibitively long times. See Figure 3 for their depiction of the comparison between passive

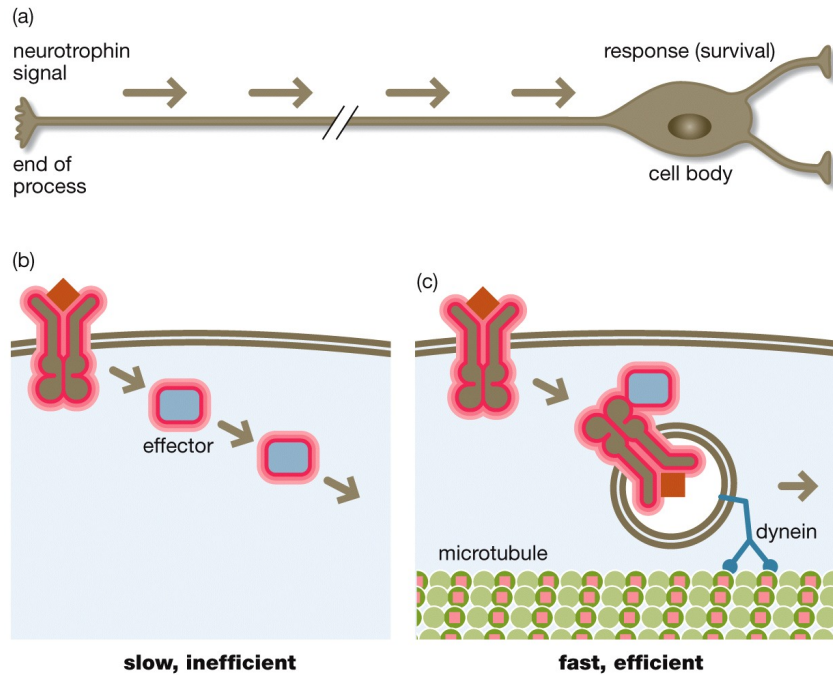


Figure 5.17 Cell Signaling (© Garland Science 2015)

Figure 3: Comparison between passive diffusion and active transport in neurons. (a) Schematic of a neuron. (b) An effector molecule is activated and then diffuses along the axon to the cell body. (c) Receptor is incorporated into a vesicle and then actively transported by a dynein molecule along a microtubule.

diffusion and active transport. Using what we have learned about diffusion, work out the time for diffusion of a protein over the 10 cm length of a neuron. Compare this to the time for a molecule to be transported actively by a motor. Do you agree with their assessment that active transport is efficient?

3. Spread the Butter Diffusion Style.

(a) In class I sketched how to derive the master equation for diffusion as a limiting process by thinking of one-dimensional line as discretized into boxes of width a , and with a jump probability $k\Delta t$. Given that at $t = 0$ we have $p(0, 0) = 1$ (i.e. all the probability is concentrated at the origin), work out

the probability after time Δt by hand. That is, figure out $p(0, \Delta t)$, $p(a, \Delta t)$ and $p(-a, \Delta t)$. Why do we not consider any points farther from the origin after a single time step?

(b) In this part of the problem, you will write a code to carry out the discrete spread the butter that you began by hand above. The goal is to work out the time evolution of the probability distribution for the initial condition when all the probability is concentrated at the origin. What you need to hand in is a plot that shows how the probability distribution changes over time.

(c) Now do the same thing as in part (b), but for the case in which the diffusion is in a finite box. The novelty here is figuring out what happens because of the boundaries. Intuitively, what will happen in the long-time limit given that you started out with a delta function at the origin? Hand in a plot of how the probability changes as a function of time. Make sure you explain what is going on in the long-time limit.

(d) One-dimensional FRAP via spread the butter. In this part of the problem, we build on what you did in the previous part to consider fluorescence-recovery-after-photobleaching for a one-dimensional cell. Consider the one-dimensional finite box with an initially uniform distribution of intensity throughout the “cell”. Now, photobleach (i.e. just set the probability to zero) a region symmetrically displaced around the origin. If the cell runs from $-L$ to L then photobleach the region from $-a$ to a . How does the recovery time depend upon the diffusion constant?